

PGP (Pretty Good Privacy) came into existence in the early 1990s and GPG (GNU Privacy Guard) that conformed to the OpenPGP standard was available by the end of the 90s.

I have used a public key only to start an SSH session on a remote computer without having to give a password. But I have relied on GPG whenever I installed packages from a Fedora repository. Keeping the private key safe is a critical part of these security processes. The fear that the signing key may have been compromised resulted in the closure of the Fedora repositories for a noticeable period of time.

One reason for the lack of applications using OpenPGP may be that it is hard to get started with them. It is important to realise that this technique is based on people trusting each other and not on a third-party certificate. Would I trust the keys more if the issuing company had been audited by, say, PWC? A transaction between two parties does not need a certificate from a third party.

Before getting into programming using GPG, let us consider the steps involved in using the public key infrastructure with an e-mail client, Evolution. We choose Evolution as it comes with GPG support. Many e-mail clients now support OpenPGP -- for example, Sylpheed. Thunderbird requires the Enigmail plug-in, which, unfortunately, was not compatible with the x64 application I was using. The default security mechanism of Thunderbird is S/MIME. Go to www.mozilla-enigmail.org/forum/viewtopic.php?f=7&t=67 for more details.

GPG and e-mail

The first step is to create your own pair of keys for your e-mail account, user@example.com. It is simple. Just give the following command:

```
gpg --gen-key
```

You will be asked a few

questions and if in doubt, just use the defaults. It is better if you give a passphrase to protect your private key, especially if others may have access to the system you are using.

You will need to send your public key to your collaborators. So, export it as a text file and e-mail it:

```
gpg -a --export user@example.com >
my_public_key.asc
```

Your friends can call you to verify that the fingerprint of the key is valid. You can find out the fingerprint by the following command:

```
gpg --fingerprint
```

You can now sign and send an e-mail to your friends. In Evolution, choose the option 'Security' on the menu bar. Select the 'PGP Sign' check box.

When you get the public key from your friends and collaborators, you will need to import it. This step is also simple:

```
gpg --import his_public_key.asc
```

GPG expects each key to be signed by a trusted entity before it is regarded as valid. So, you will need to sign the key you have just imported as follows—assuming that your friend's e-mail address is friend@example.com:

```
gpg --sign-key friend@example.com
```

Now, you can encrypt the e-mail you are sending to your friend. When composing an e-mail, choose the 'Security' option from the menu bar and select the 'PGP Encrypt' check box. You can encrypt and sign the e-mail by selecting both the sign and the encrypt check boxes.

If you have received an encrypted and signed e-mail from your friend, Evolution will display it as usual, except that there will be a message at the bottom of the e-mail informing you that it had a valid signature and was encrypted.

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